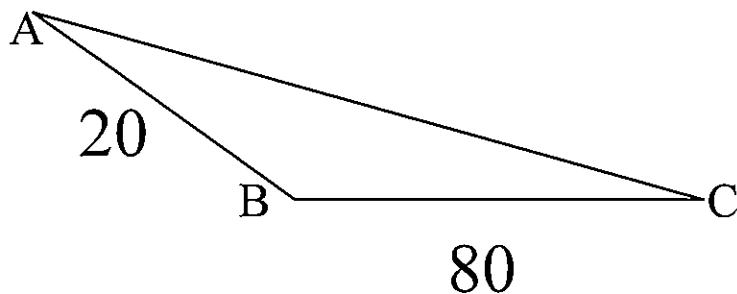


1. $\begin{bmatrix} 3 & 2 \\ 4 & 6 \end{bmatrix} \begin{bmatrix} 5 \\ -1 \end{bmatrix} = \begin{bmatrix} k \\ w \end{bmatrix}$. Find the value of $(2k + 3w)$.
2. Given the set: $\{\log_2(16), \log_3(16), \log_4(16), \log_5(16)\}$. If one of the 4 members of the set is drawn at random, find the probability that the member drawn could represent a positive integer. Express your answer as a common fraction reduced to lowest terms.
3. Find the ordered pair that represents the sum of the following two vectors: $(-5, 6)$ and $(17, 7)$.
4. Let n be a positive integer. For $n \geq 1$, $x_{(n+2)} = x_{(n)} + x_{(n+1)}$. If $x_1 = 3$ and $x_2 = 1$, find the value of x_{10} .
5. Matrix $[A]$ contains 23 rows and 13 columns. Matrix $[B]$ contains 13 rows and 92 columns. Find the number of rows in the product matrix $[A][B]$.
6. In $\triangle ABC$ as with side-lengths as shown, $\angle ABC = 120^\circ$. Find the exact length of $\overline{AC}$.



7. A woman is sitting in the stands behind one end zone of a football field. She observes, in the same vertical plane with her position, two players standing on their own goal lines, exactly 100 yards apart. Looking at the spots on the ground where they are standing, a player at one goal line is at an angle of depression of 15° and a player at the other goal line is at an angle of depression of 8° . Assuming the football field is horizontal, find the **number of feet** in the vertical height of the woman's eye above the horizontal plane of the football field. Express your answer as a **decimal** rounded to the nearest hundredth of a foot.